

Wikipedia

Annotating a corpus against Wikipedia articles

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Wikipedia: what is it here

Wikipedia needs no introduction as an encyclopaedia. What matters for annotation is that it is also a machine-readable resource: every article has a stable title and a stable address, and Wikipedia can be asked, for a whole list of names at once, whether an article exists for each of them, what its canonical title is, and where an alternative spelling redirects.

How Wikipedia differs from DBpedia and YAGO

DBpedia and YAGO both *extract* structured knowledge from Wikipedia and Wikidata and organize it with an ontology — DBpedia its own, YAGO the schema.org classes. Annotating with either means choosing an ontology class and tagging the words that belong to it.

Wikipedia has no ontology, so annotating against Wikipedia asks a simpler question: *does this name have an article?* If it does, the word is linked to it. There is no class to select, which is why the Ontology class and search fields stay greyed out when Wikipedia is chosen.

- **Use Wikipedia** when your corpus names people, places and organizations that are too local or too minor to have entered an ontology — the normal case for nineteenth-century material.
- **Use DBpedia or YAGO** when you want to count or filter BY CATEGORY: every Person, every Organization. Only an ontology can answer that.
- **An ontology also disambiguates.** Asking YAGO for a Place rules out the film and the ship of the same name; Wikipedia alone cannot.

What gets annotated: your choice

Wikipedia has no ontology classes, so the **Ontology class** dropdown offers instead a choice of WHAT to annotate:

- **Proper nouns only (people, places, organizations)** — the default. Single words and multi-word sequences alike, so Jim Cobb and The Atlanta Constitution are each treated as one name.

- **All content words (also common nouns)** — nouns, adjectives and adverbs as well as names.

Neither setting annotates verbs. In fact **no annotator in the NLP Suite tags verbs**: DBpedia and YAGO exclude them too, along with determiners, prepositions, pronouns and auxiliaries.

Which to choose

The default exists because Wikipedia carries articles for common nouns as readily as for names — *sheriff*, *lynching* and *jail* all have one. DBpedia and YAGO keep their annotation manageable through the ontology class you select; Wikipedia offers nothing equivalent, so the wider setting links a large share of the words in a document, and a page on which almost everything is a link tells the reader very little.

Choose **proper nouns only** for ENTITY linking: which name refers to which person, place or organization. This is what a knowledge graph is usually wanted for, and it is the setting to use when you intend to count or map the people and places a corpus names.

Choose **all content words** when the question is about CONCEPTS rather than names — whether *lynching*, *mob* or *due process* themselves have an encyclopaedic presence, and which of a period's vocabulary the encyclopaedia records. Expect many more links, and expect to read them critically: a common noun links to the general article, so *jail* arrives at the article on *Prison*.

The point worth keeping in view is that this is a *scholarly* decision, not a technical one. Neither setting is more correct; they answer different questions, and the NLP Suite leaves the choice to you rather than making it on your behalf.

How a name is resolved

Four things happen between the word in your document and the article it links to.

- **Titles and honorifics are set aside for the lookup.** Newspapers write "Sheriff Jim Cobb" and "Governor Hugh Dorsey"; Wikipedia files the article under the bare name, so the honorific is dropped before asking. The annotation still covers the phrase as it appears in your text.
- **Redirects are followed.** "The Atlanta Constitution" resolves to the article "The Atlanta Journal-Constitution".
- **Disambiguation pages are dropped.** "Macon" leads to a page listing nine different places of that name, which identifies nothing, so no link is made.
- **Month and weekday names are skipped.** In a corpus of dated newspaper articles, every "May" would otherwise link to the article about the month.

Annotating with Wikipedia in the NLP Suite

Wikipedia is annotated from the same GUI as DBpedia and YAGO (the Knowledge graphs annotator). Select Wikipedia from the Knowledge bases dropdown, use the Ontology class dropdown to choose what you want annotated (see above), pick a colour, and press RUN. The sub-class and search fields stay disabled: they belong to the ontology-based annotators.

- **INPUT:** one or more .txt files.

- **OUTPUT:** one annotated .html file per document, with each linked name in the colour you chose, plus a csv listing every annotated word with its article address, the sentence it came from and the document it came from, and a frequency chart of the entities found.

An internet connection is required: the articles are looked up as the corpus is processed. Names are sent in batches, so a corpus of ordinary size costs few requests, but a very large corpus will take time.

Cautions

A bare, ambiguous place name can land on the wrong article, because Wikipedia's own redirect decides where it goes. *Savannah* resolves to *Savanna*, the grassland, and not to the city in Georgia. This is the clearest argument for using DBpedia or YAGO when an ontology class could settle the question.

Coverage is uneven in exactly the way that matters for historical work: national figures and large cities are well covered, while local nineteenth-century people and places often have no article at all. The absence of a link is therefore not evidence that a name is unimportant — only that Wikipedia does not carry it.

As with every knowledge base, the annotation is a judgment made by software. Treat the output as a first pass to be inspected, not as a finished classification.

Further readings — more TIPS files

TIPS_NLP_YAGO.pdf · TIPS_NLP_Annotator DBpedia.pdf · TIPS_NLP_Annotator.pdf ·
TIPS_NLP_W3C OWL RDF SPARQL.pdf

Links: <https://en.wikipedia.org/> · https://www.mediawiki.org/wiki/API:Main_page